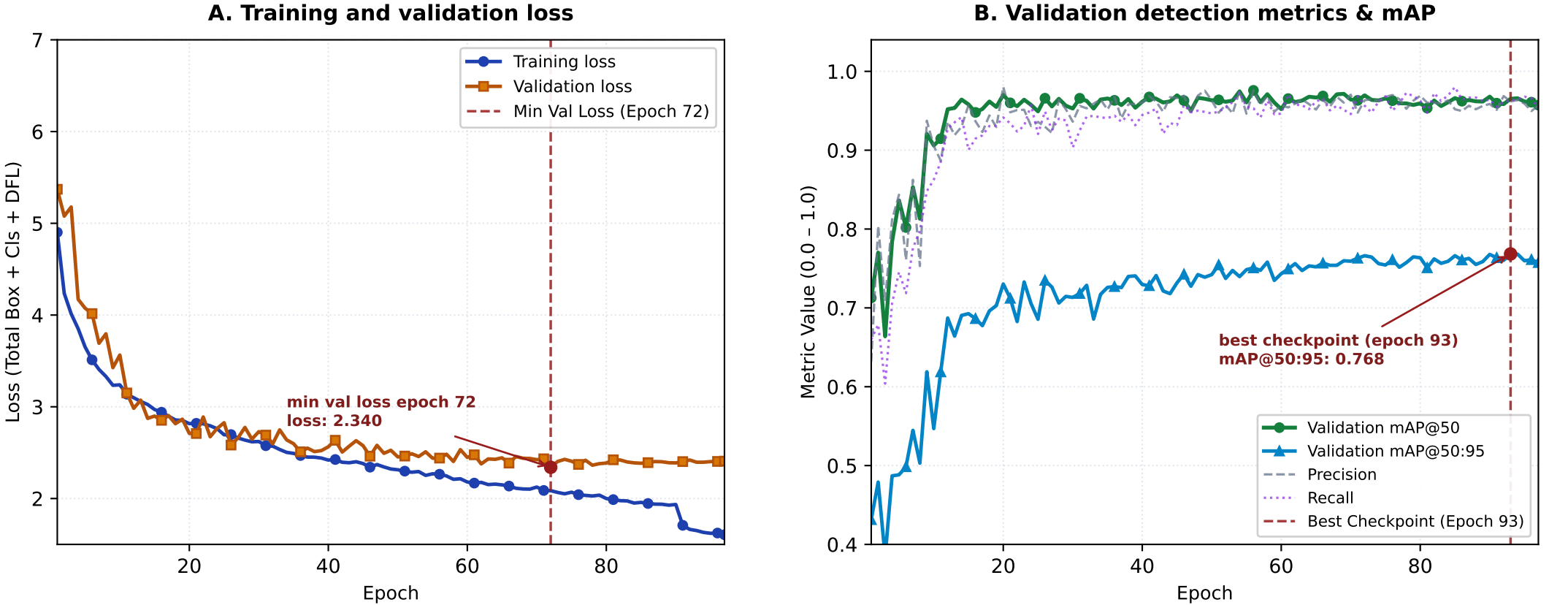
Figure 7. YOLOv8n Baseline Training and Validation Learning Curves

YOLOv8n Learning Curves — Baseline Run 39f402dfe460 (3,006,038 parameters, 100 epochs on MPS)



Training and validation learning curves for YOLOv8n (3.01M parameters) across 100 epochs: (A) Total loss (CIoU bounding box + classification + distribution focal loss) illustrating steady monotonic reduction in training loss while validation loss steadily drops and stabilizes near 2.340 (epoch 72), and (B) Validation performance trajectory showing rapid convergence of mAP@50 and mAP@50:95. Diagnosis: The parallel downward trend and absence of validation loss divergence indicate clean convergence without severe overfitting; the best model checkpoint is restored at epoch 93 (mAP@50: 0.9652, mAP@50:95: 0.7684).